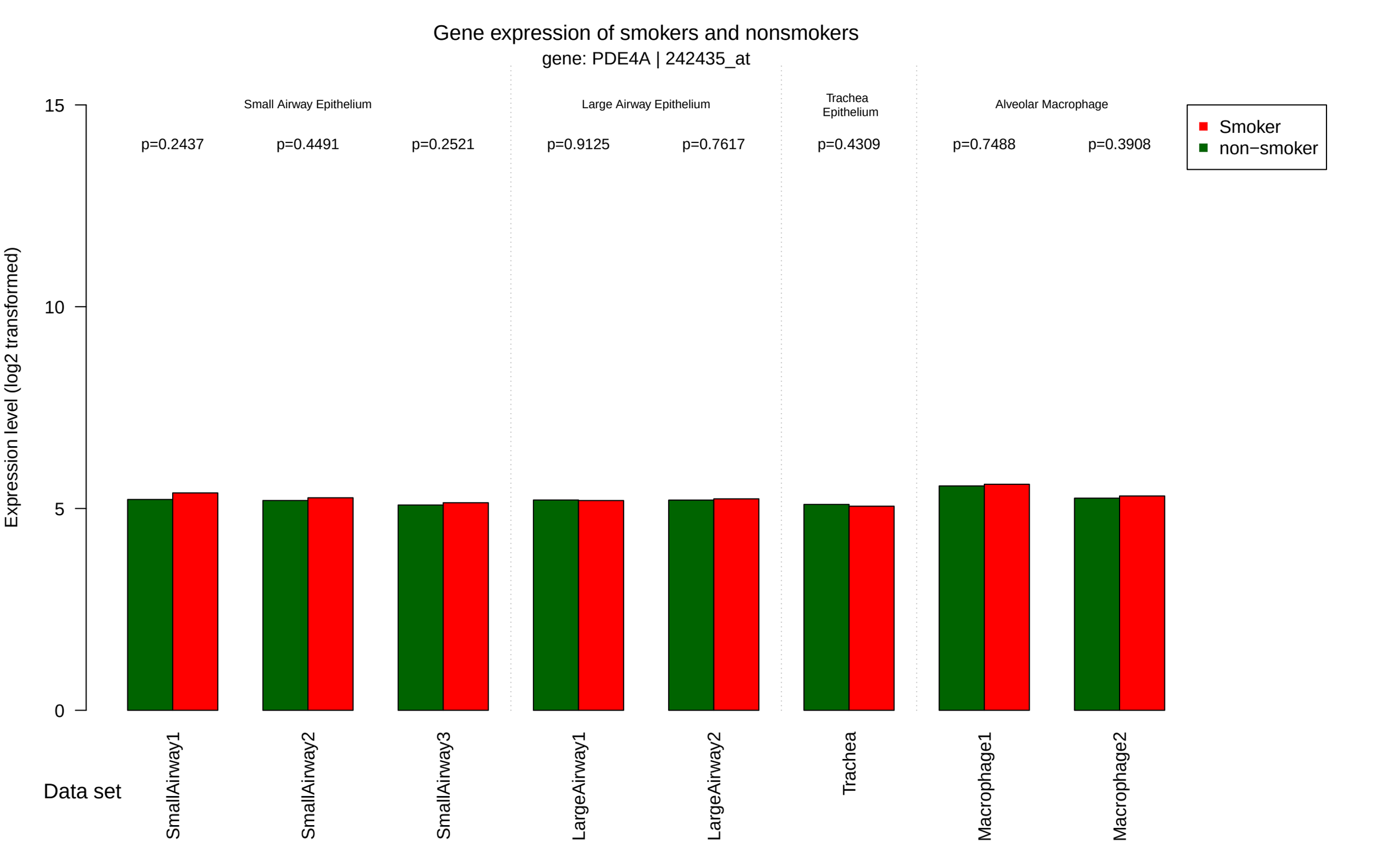
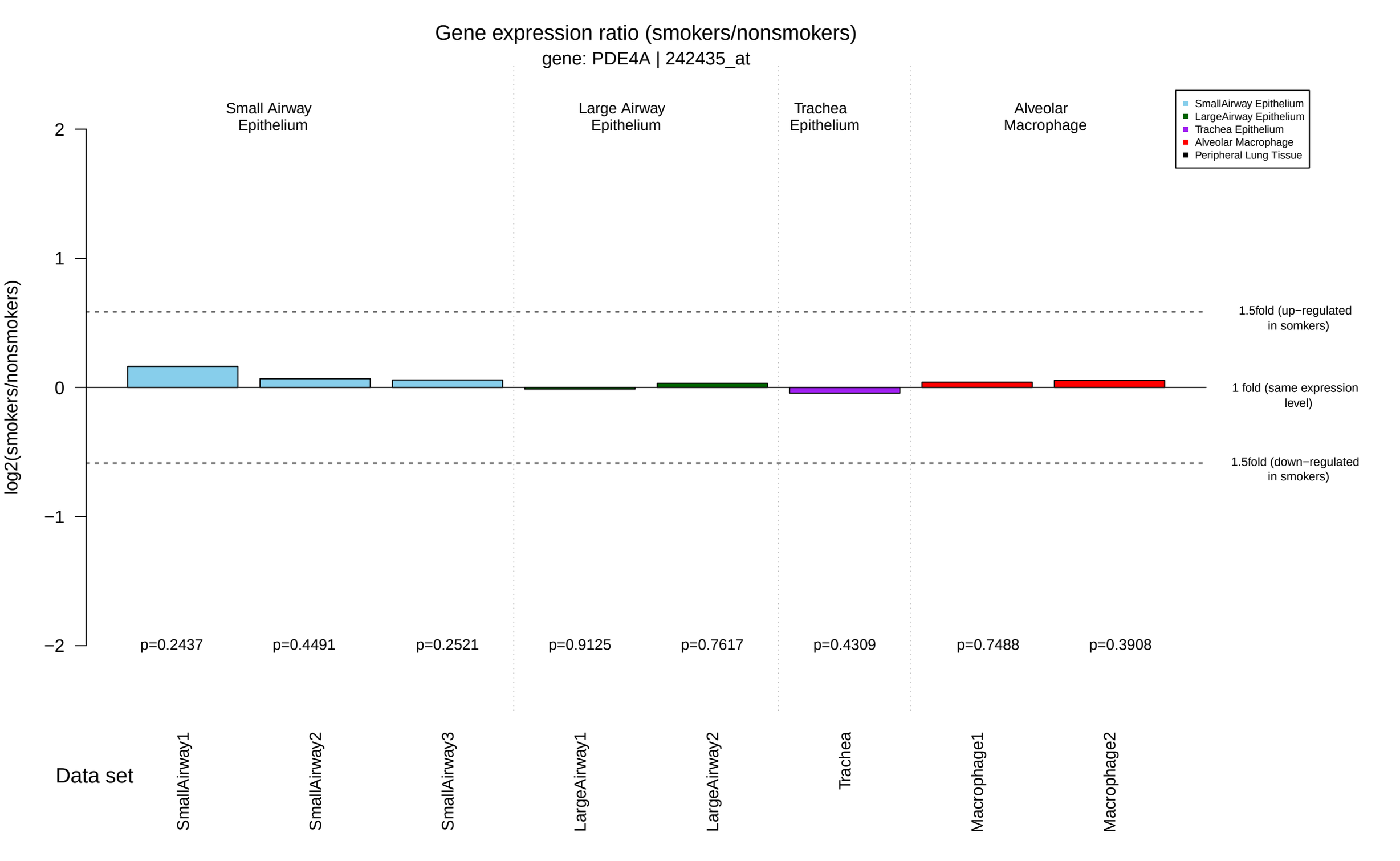
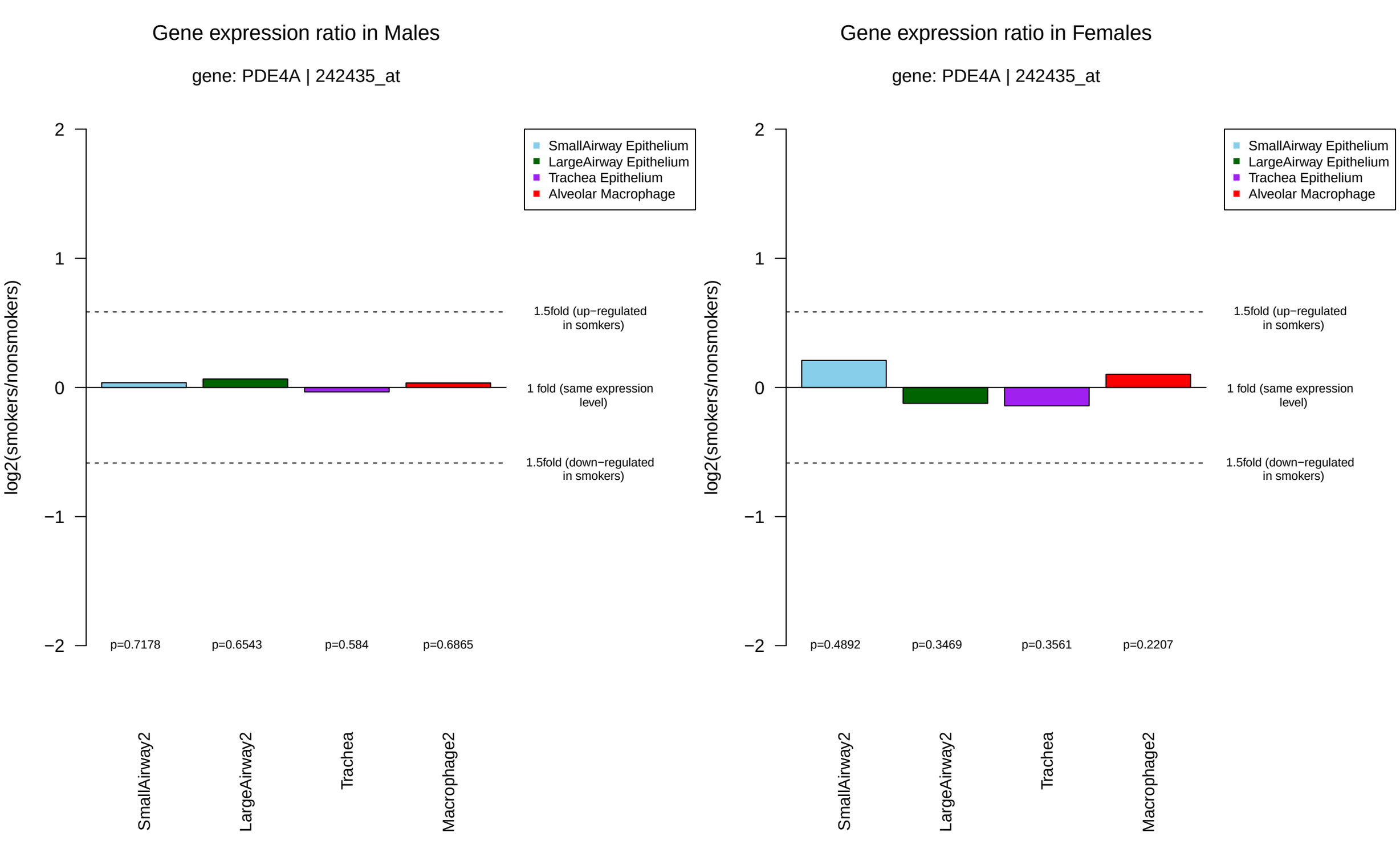
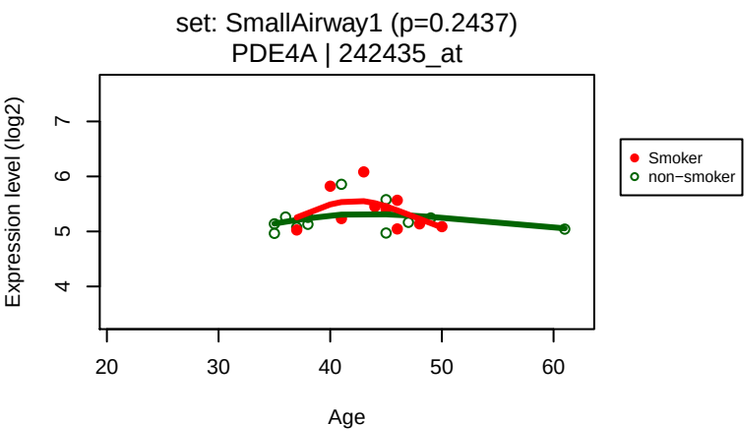
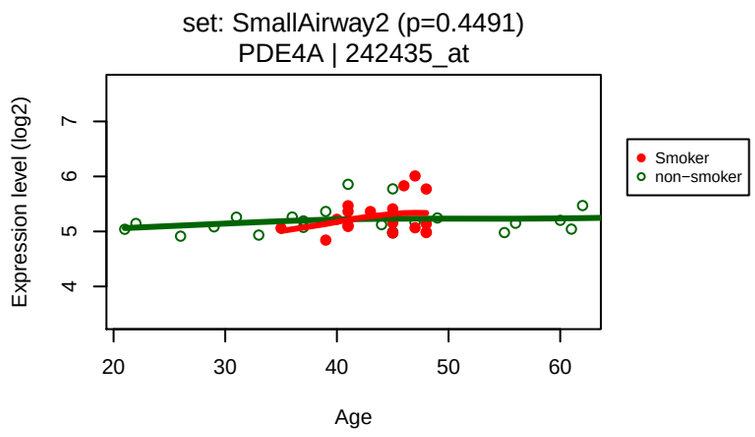
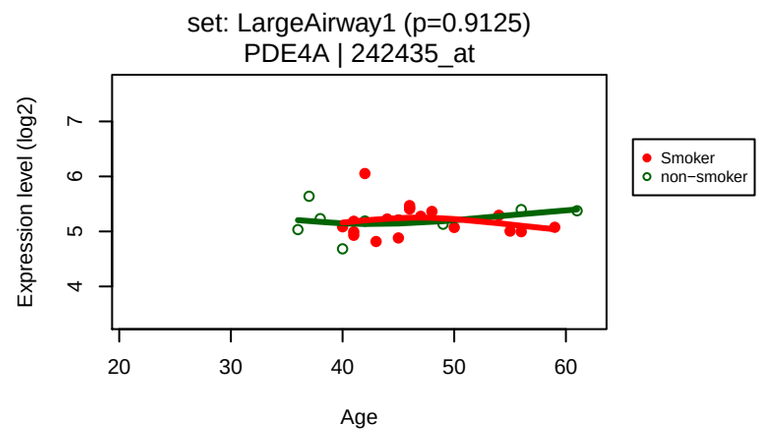

Gene expression of smokers and nonsmokers
gene: PDE4A | 242435_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2437
p=0.4491
p=0.2521
p=0.9125
p=0.7617
p=0.4309
p=0.7488
p=0.3908
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

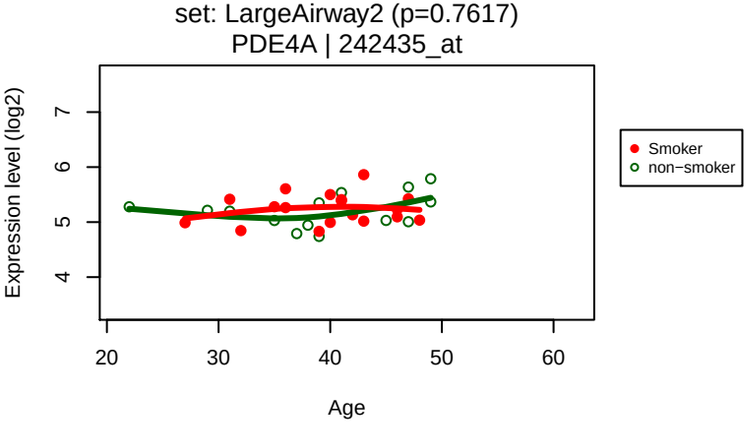

Gene expression ratio (smokers/nonsmokers)
gene: PDE4A | 242435_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2437
p=0.4491
p=0.2521
p=0.9125
p=0.7617
p=0.4309
p=0.7488
p=0.3908
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

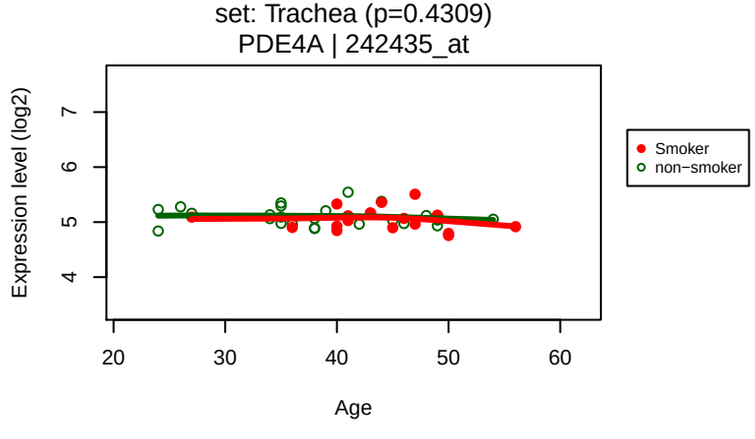

Gene expression ratio in Males
gene: PDE4A | 242435_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7178
p=0.6543
p=0.584
p=0.6865
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE4A | 242435_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4892
p=0.3469
p=0.3561
p=0.2207
SmallAirway2
LargeAirway2
Trachea
Macrophage2

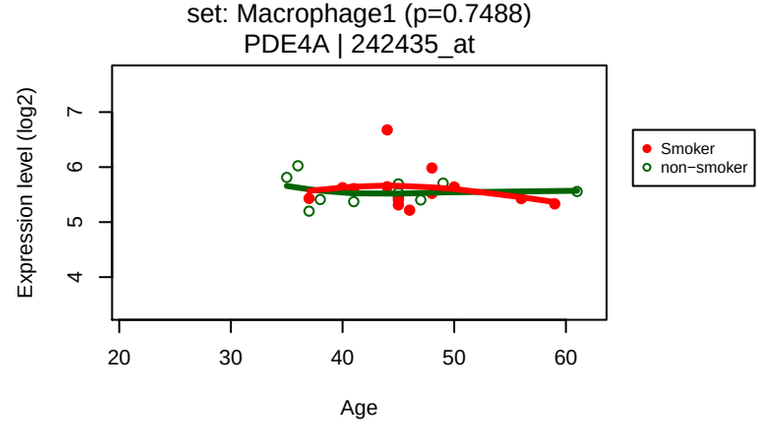

set: SmallAirway1 (p=0.2437)
PDE4A | 242435_at
Expression level (log2)
Age
Smoker
non−smoker

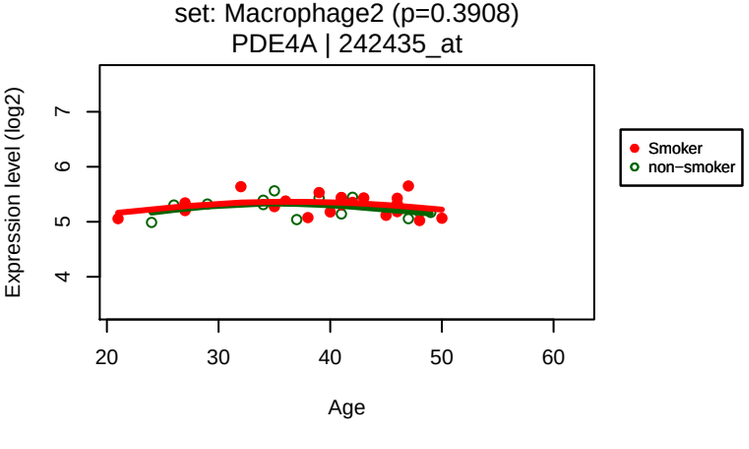

set: SmallAirway2 (p=0.4491)
PDE4A | 242435_at
Expression level (log2)
Age
Smoker
non−smoker

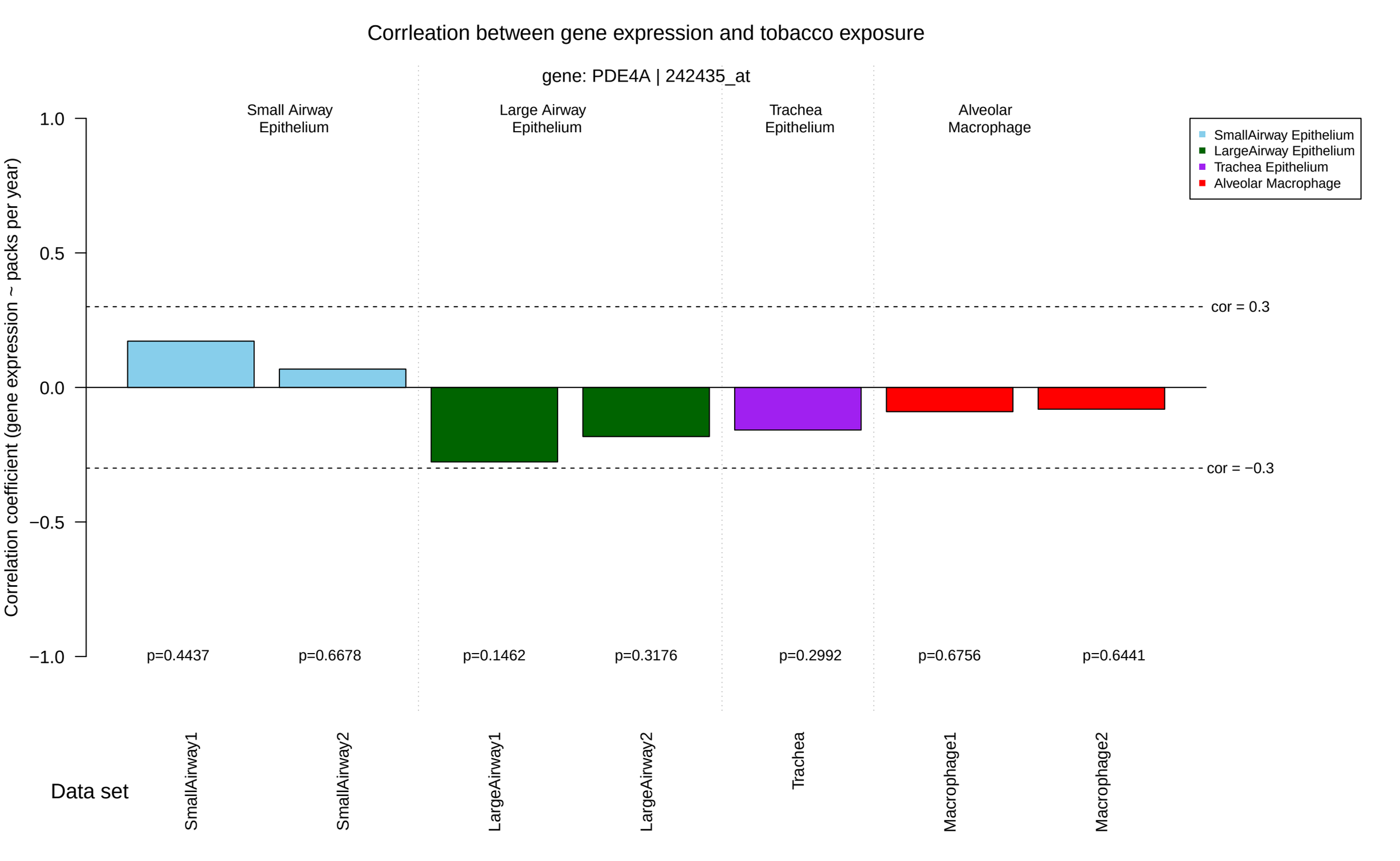

set: LargeAirway1 (p=0.9125)
PDE4A | 242435_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7617)
PDE4A | 242435_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4309)
PDE4A | 242435_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7488)
PDE4A | 242435_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3908)
PDE4A | 242435_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE4A | 242435_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4437
p=0.6678
p=0.1462
p=0.3176
p=0.2992
p=0.6756
p=0.6441
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2